

|                                       |                                                                                                                                                                                                                                                       |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Project Name</b>                   | Project 2 - NIM - Automated Thematic Analysis                                                                                                                                                                                                         |
| <b>Online team meeting</b>            | <a href="https://fau.zoom-x.de/j/64229233763?pwd=MboXckMTe5aVIBRlWvO71OggjXJ7zb.1">https://fau.zoom-x.de/j/64229233763?pwd=MboXckMTe5aVIBRlWvO71OggjXJ7zb.1</a>                                                                                       |
| <b>Production system (if any)</b>     | ...                                                                                                                                                                                                                                                   |
| <b>Test system (if any)</b>           | ...                                                                                                                                                                                                                                                   |
| <b>GitHub repository</b>              | <a href="https://github.com/amosproj/amos2026ss02-automated-thematic-analysis">https://github.com/amosproj/amos2026ss02-automated-thematic-analysis</a>                                                                                               |
| <b>GitHub feature board</b>           | <a href="https://github.com/orgs/amosproj/projects/98">https://github.com/orgs/amosproj/projects/98</a>                                                                                                                                               |
| <b>GitHub imp-squared backlog</b>     | <a href="https://github.com/orgs/amosproj/projects/102">https://github.com/orgs/amosproj/projects/102</a>                                                                                                                                             |
| <b>Position of T-shirt logo</b>       | Upper Torso Right Small                                                                                                                                                                                                                               |
| <b>Link to logo for black T-shirt</b> | <a href="https://github.com/amosproj/amos2026ss02-automated-thematic-analysis/blob/main/Deliverables/sprint-01/team-logo.svg">https://github.com/amosproj/amos2026ss02-automated-thematic-analysis/blob/main/Deliverables/sprint-01/team-logo.svg</a> |
| <b>Link to logo for white T-shirt</b> | <a href="https://github.com/amosproj/amos2026ss02-automated-thematic-analysis/blob/main/Deliverables/sprint-01/team-logo.svg">https://github.com/amosproj/amos2026ss02-automated-thematic-analysis/blob/main/Deliverables/sprint-01/team-logo.svg</a> |
| <b>Additional materials</b>           | ...                                                                                                                                                                                                                                                   |
| <b>Team mailing list</b>              | oss-amos-proj2@lists.fau.de (Emails sent here get quarantined by the TU Berlin Spam Filter)                                                                                                                                                           |

| Last Name   | First Name | GitHub User Name        | Email Address                             |
|-------------|------------|-------------------------|-------------------------------------------|
| Kahvecioglu | Selin      | slnkahveci              | kahvecioglu@campus.tu-berlin.de           |
| de Sa Filho | Fabiano    | fdesafilho              | fabiano.de.sa@fau.de                      |
| Ammann      | Paul       | Wecoreator              | p.ammann@campus.tu-berlin.de              |
| Schmidt     | Alexander  | alexanderschmi          | a.schmidt.2@tu-berlin.de                  |
| Wendlandt   | Julian     | julianlucawendlandt-jpg | julian.luca.wendlandt@campus.tu-berlin.de |
| Müermann    | Ole        | ClimbTheGrind           | ole.mueermann@fau.de                      |
| Klarer      | Paul       | paulKlarer              | paul.linus.klarer@fau.de                  |
| Mantsch     | Christoph  | Christoph-Mantsch       | christoph.cm.mantsch@fau.de               |
| Annus       | Mohammad   | annus3                  | mohammad.annus@fau.de                     |
| Rasul       | Hamid      | hrasul                  | hamid.rasul@fau.de                        |
|             |            |                         |                                           |
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|             |            |                         |                                           |

| #                                                                                                                                                                       | Meeting Day | Product Owner       |                     | Software Developer | Release Manager   | Scrum Master      | Comment       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------------|---------------------|--------------------|-------------------|-------------------|---------------|
|                                                                                                                                                                         |             | Review              | Planning            |                    |                   |                   |               |
| 1                                                                                                                                                                       | 2026-04-15  | Alexander Schmidt   | Fabiano de Sa Filho | Everyone else      | N/A               | Christoph Mantsch |               |
| 2                                                                                                                                                                       | 2026-04-22  | Alexander Schmidt   | Fabiano de Sa Filho | Everyone else      | Paul Klarer       | Christoph Mantsch |               |
| 3                                                                                                                                                                       | 2026-04-29  | Fabiano de Sa Filho | Alexander Schmidt   | Everyone else      | Paul Klarer       | Christoph Mantsch |               |
| 4                                                                                                                                                                       | 2026-05-06  | Alexander Schmidt   | Fabiano de Sa Filho | Everyone else      | Selin Kahvecioglu | Christoph Mantsch |               |
| 5                                                                                                                                                                       | 2026-05-13  | Fabiano de Sa Filho | Alexander Schmidt   | Everyone else      | Ole Müermann      | Christoph Mantsch |               |
| 6                                                                                                                                                                       | 2026-05-20  | Alexander Schmidt   | Fabiano de Sa Filho | Everyone else      | Ole Müermann      | Christoph Mantsch |               |
| 7                                                                                                                                                                       | 2026-05-27  | Fabiano de Sa Filho | Alexander Schmidt   | Everyone else      | Hamid Rasul       | Christoph Mantsch | Mid-term due  |
| 8                                                                                                                                                                       | 2026-06-03  | Alexander Schmidt   | Fabiano de Sa Filho | Everyone else      | Selin Kahvecioglu | Christoph Mantsch |               |
| 9                                                                                                                                                                       | 2026-06-10  | Fabiano de Sa Filho | Alexander Schmidt   | Everyone else      | Hamid Rasul       | Christoph Mantsch |               |
| 10                                                                                                                                                                      | 2026-06-17  | Alexander Schmidt   | Fabiano de Sa Filho | Everyone else      | Paul Ammann       | Christoph Mantsch |               |
| 11                                                                                                                                                                      | 2026-06-24  | Fabiano de Sa Filho | Alexander Schmidt   | Everyone else      | Mohammad Annus    | Christoph Mantsch |               |
| 12                                                                                                                                                                      | 2026-07-01  | Alexander Schmidt   | Fabiano de Sa Filho | Everyone else      | Mohammad Annus    | Christoph Mantsch |               |
| 13                                                                                                                                                                      | 2026-07-08  | Fabiano de Sa Filho | Alexander Schmidt   | Everyone else      | Paul Ammann       | Christoph Mantsch |               |
| 14                                                                                                                                                                      | 2026-07-15  | Alexander Schmidt   | Fabiano de Sa Filho | Everyone else      | Julian Wendlandt  | Christoph Mantsch | Demo day!     |
| 15                                                                                                                                                                      | 2026-07-22  | Fabiano de Sa Filho | Alexander Schmidt   | Everyone else      | Julian Wendlandt  | Christoph Mantsch | Retrospective |
| Product owners, software developers, and Scrum Master are set and ideally don't change over time; the critical part is the Release Manager role you need to define here |             |                     |                     |                    |                   |                   |               |

|                                |                                                                                                                                                                                |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Goals</b>                   | Having Fun                                                                                                                                                                     |
|                                | Satisfy NIM                                                                                                                                                                    |
|                                | Program an automated thematic analysis of interview transcripts                                                                                                                |
| <b>Meeting norms</b>           | Joining the meeting is mandatory                                                                                                                                               |
|                                | Always be on time (up to 10 min late is fine if short message on discord)                                                                                                      |
|                                | Follow the timeline                                                                                                                                                            |
|                                | Take notes in each meeting                                                                                                                                                     |
|                                | Happiness Index after Sprint retrospective                                                                                                                                     |
|                                | Each meeting takes 1:30 hours                                                                                                                                                  |
| <b>Working norms</b>           | Tell us if/when you are working on a task                                                                                                                                      |
|                                | We encourage pair programming                                                                                                                                                  |
| <b>Coordination norms</b>      | Self assign issues you are opening                                                                                                                                             |
|                                | Do not force push                                                                                                                                                              |
|                                | Use PEP 8 as style (configure your IDEs)                                                                                                                                       |
|                                | <a href="https://www.atlassian.com/git/tutorials/comparing-workflows/gitflow-workflow">Follow https://www.atlassian.com/git/tutorials/comparing-workflows/gitflow-workflow</a> |
|                                | Once every workday check discord                                                                                                                                               |
|                                | If someone is blocked share with the team                                                                                                                                      |
| <b>Communication norms</b>     | Criticism should be productive                                                                                                                                                 |
|                                | Language should be respectful at all time (written and oral)                                                                                                                   |
| <b>Consideration norms</b>     | Discuss disagreement openly                                                                                                                                                    |
|                                | Follow the NDA and data privacy                                                                                                                                                |
|                                | Try not to induce bias in our project                                                                                                                                          |
|                                |                                                                                                                                                                                |
| <b>Cont. improvement norms</b> | Finish work until tuesday evening 20:00 PM                                                                                                                                     |
|                                | At the end of the meeting discuss happiness score / what could be done better / what was good                                                                                  |
| <b>Rewards</b>                 | Have a happy meeting                                                                                                                                                           |
|                                |                                                                                                                                                                                |
| <b>Sanctions</b>               | Write a poem saying you are sorry                                                                                                                                              |
|                                | After multiple or serious offenses escalate to prof                                                                                                                            |
| <b>Signatures</b>              |                                                                                                                                                                                |
|                                |                                                                                                                                                                                |
| Scrum Master                   | Christoph Mantsch                                                                                                                                                              |
| Product owner                  | Alexander Schmidt                                                                                                                                                              |
| Product owner                  | Fabiano de Sa Filho                                                                                                                                                            |
| Software developer             | Ole Müermann                                                                                                                                                                   |
| Software developer             | Julian Wendlandt                                                                                                                                                               |
| Software developer             | Hamid Rasul                                                                                                                                                                    |
| Software developer             | Paul Klarer                                                                                                                                                                    |

|                    |                   |
|--------------------|-------------------|
| Software developer | Paul Ammann       |
| Software developer | Selin Kahvecioglu |
| Software developer | Mohammad Annus    |

| Product Vision                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Project Mission                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LLMs are a fantastic tool to interview research subjects at scale, which previously had to be done by human researchers. This method generates unstructured interview dialogues which must then be processed and analysed manually. The goal of this project is to use LLMs to also automate the analysis part, enabling policy researchers to feed their interview corpus to our tool and obtain not only the thematic analysis, but also relevant insights derived from the demographic data of the respondents. This way, we seek to bridge the gap between quantitative and qualitative policy research. | To engineer a secure, open-source tool that ingests thousands of subject interviews about a policy topic and produces an accurate thematic analysis breakdown. The core is a multi-stage LLM pipeline that extracts quotes, codes them into themes, clusters and ranks those themes, and cross-references them against respondent demographic dimensions. Results are surfaced through a lightweight frontend and exportable in structured formats. We also provide direct reference back to the corpus, so researchers can see which interviews most strongly manifest a particular theme or demographic combination. |

| Term                       | Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Corpus / Dataset           | The LLM-led interviews on the economic impact of AI provided by the Industry Partner. Format: JSON                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Demographic Data           | Categorical variables describing each subject in the Corpus; has been partially sanitized by Industry partner, but must be handled with care. Absolutely should not be downloaded or uploaded to third parties. Format: CSV. Type: Numerical or Categorical Variables                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Thematic Analysis          | The qualitative method of identifying and analyzing patterns of meaning (themes) in a dataset. These themes should be followed by a short description and some quantitative measures of prevalence in the dataset.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Analysis Run               | A single execution of the thematic analysis process based on a defined corpus, query, coding mode, and codebook. An analysis run produces outputs such as themes, tagged passages, and aggregated results.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Codebook                   | A structured collection of themes/codes used during thematic analysis. A codebook typically contains: <ul style="list-style-type: none"> <li>- code/theme name</li> <li>- description</li> <li>- optional inclusion/exclusion criteria</li> <li>- optional examples</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Interview                  | A textual dialogue between a research subject and an LLM about a research topic. It is provided by the industry partner in our case, and the interviews are about the Economic Impacts of AI in the job market.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Demographic Data Variables | Can be of two types: (1) Numerical: Continuous variable like age, income. (2) Categorical: Classifies respondents into a category, such as gender, age bracket, level of schooling.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Coding Mode                | The method by which the system determines and applies themes. The project currently includes three coding modes: <ul style="list-style-type: none"> <li>- Fully automatic: The system consumes the corpus and produces the entire codebook without any human supervision, generating insights from the demographic data.</li> <li>- Semi-automatic: The system uses the corpus to produce a codebook, but allows the user to add, remove, and edit any of the themes. The system then proceeds with the analyses using this edited codebook and receives natural language queries from the user, but does not modify the user-provided codebook in any way.</li> <li>- User-instructed: The system bypasses the codebook generation phase, and goes straight in to the natural language queries with the user provided codebook.</li> </ul> |
| Corpus View                | The product view that presents an aggregated overview of themes across the full corpus, including frequencies, quotes, and demographic breakdowns.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Interview View             | The product view showing one full transcript, its highlighted themes, and the associated demographic information of the interviewee.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Tag                        | A thematic assignment attached to a specific passage in a transcript.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Query                      | A natural-language research question entered by the researcher to guide the thematic analysis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Theme                      | A recurring pattern of meaning relevant to the research question, identified in one or more interviews.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Qualitative Analysis       | A research method focused on interpreting non-numeric data, such as interview transcripts, to identify patterns, meanings, and themes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Code                       | A short label, usually 5–12 words, that captures the central meaning of a quote or text chunk; it also has a description clarifying its scope.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

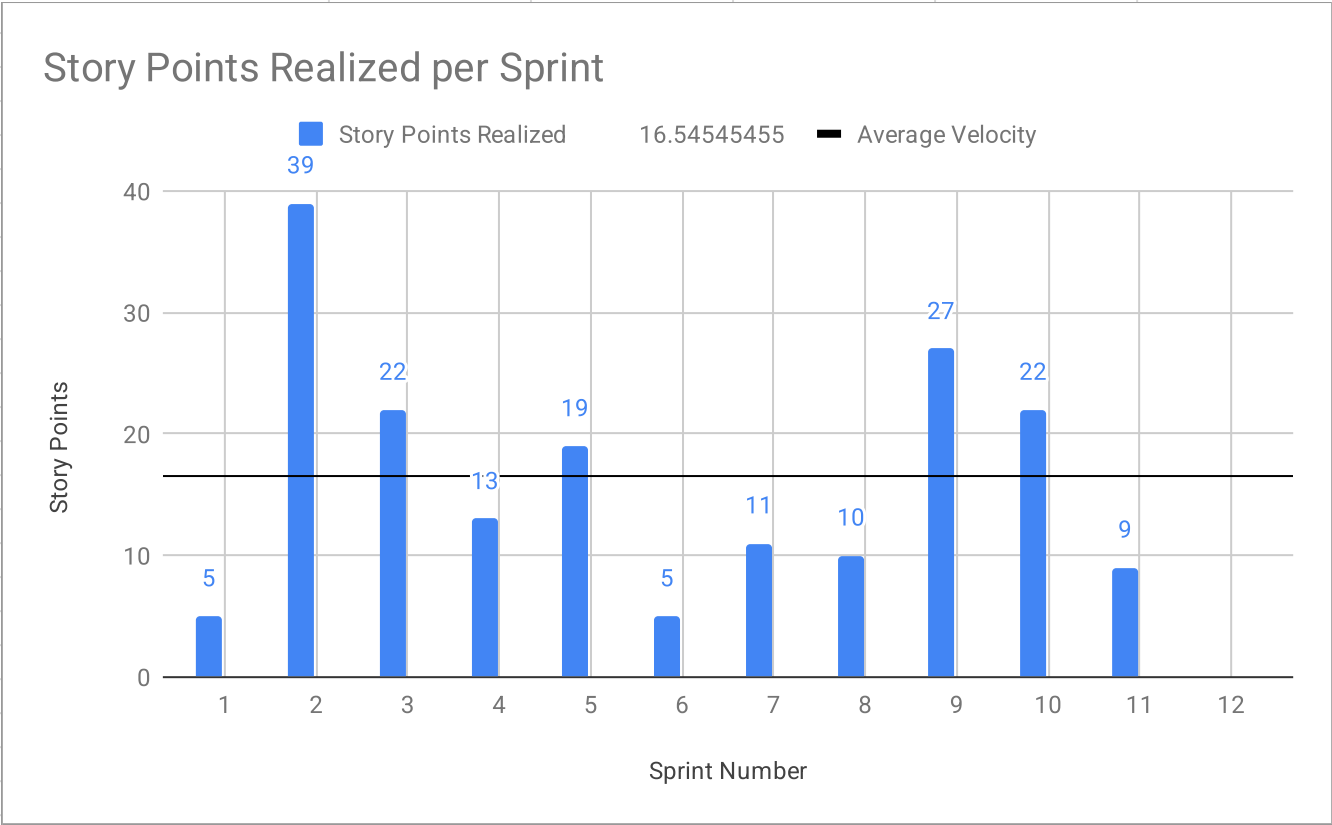
| Sprint # | Sprint goal                                                                                                                                                                                                      |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | Meeting with industry partner, designing the logo, setting up the project                                                                                                                                        |
| 2        | Ingesting the interview dataset provided by the industry partner, working on the Architecture Description document, and delivering a first prototype that successfully passes text through a basic LLM endpoint. |
| 3        | Creating a basic demo to be shown to Dr. Kaiser. Starting with simple views, database and llm communication                                                                                                      |
| 4        | Creating the first version of the frontend, finishing up the basic demo of the backend. Adding a Readme and writting a proper Build script.                                                                      |
| 5        | Closing up all features from previous sprints, finalizing the frontend so that we have a working demo for Industry Partner.                                                                                      |
| 6        | Working on the thematic analysis logic for the mid project release                                                                                                                                               |
| 7        | Finishing work of Sprint 6. Refactoring and Optimization                                                                                                                                                         |
| 8        | Start on Apply codebook to Corpus                                                                                                                                                                                |
| 9        | Finish Apply Codeboook to Corpus and other minor issues                                                                                                                                                          |
| 10       | Focus on displaying all of the transcripts for a particular theme and adding external API Support. Minor fixes requested by Dr.Kaiser                                                                            |
| 11       | Finish Analysis Visualization, Integrate incremental Codebook Generation and Application                                                                                                                         |
| 12       |                                                                                                                                                                                                                  |
| 13       |                                                                                                                                                                                                                  |
| 14       |                                                                                                                                                                                                                  |
| 15       |                                                                                                                                                                                                                  |
|          |                                                                                                                                                                                                                  |
|          |                                                                                                                                                                                                                  |
|          |                                                                                                                                                                                                                  |



| Sprint # | Story Points Realized                                                      | Story Points Planned | Average Planned | Average Realized |
|----------|----------------------------------------------------------------------------|----------------------|-----------------|------------------|
| 1        | 5                                                                          | 5                    | 15              | 16.54545455      |
| 2        | 39                                                                         | 19                   | 15              | 16.54545455      |
| 3        | 22                                                                         | 22                   | 15              | 16.54545455      |
| 4        | 13                                                                         | 10                   | 15              | 16.54545455      |
| 5        | 19                                                                         | 19                   | 15              | 16.54545455      |
| 6        | 5                                                                          | 5                    | 15              | 16.54545455      |
| 7        | 11                                                                         | 9                    | 15              | 16.54545455      |
| 8        | 10                                                                         | 12                   | 15              | 16.54545455      |
| 9        | 27                                                                         | 27                   | 15              | 16.54545455      |
| 10       | 22                                                                         | 28                   | 15              | 16.54545455      |
| 11       | 9                                                                          | 9                    | 15              | 16.54545455      |
| 12       |                                                                            |                      | 15              | 16.54545455      |
| 13       |                                                                            |                      | 15              | 16.54545455      |
| 14       |                                                                            |                      | 15              | 16.54545455      |
| 15       |                                                                            |                      | 15              | 16.54545455      |
|          | PLEASE CREATE THE VELOCITY CHART ON A NEW TAB USING THE DATA FROM THIS TAB |                      |                 |                  |
|          |                                                                            |                      |                 |                  |

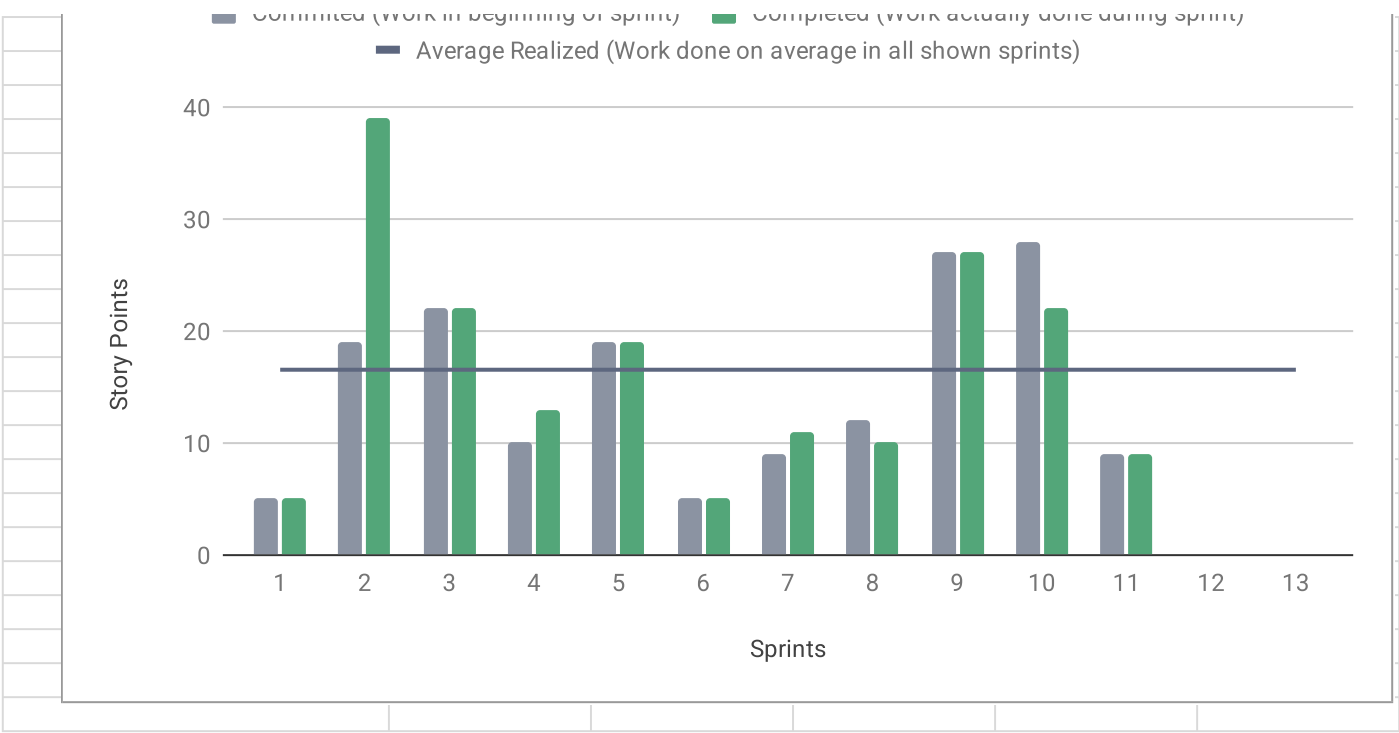
| Sprint          | Goal | Feature Name                                                                                                                                                                         | Est. size | Est. remaining | Real size | Real remaining |
|-----------------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------|-----------|----------------|
| <b>Release</b>  |      |                                                                                                                                                                                      |           |                |           |                |
| <b>Total</b>    |      |                                                                                                                                                                                      | 89        |                | 98        |                |
| <b>Sprints</b>  |      |                                                                                                                                                                                      |           |                |           |                |
| 0               |      |                                                                                                                                                                                      |           |                |           |                |
| 1               |      | Meeting with industry partner, designing the logo, setting up the project                                                                                                            | 5         | 84             | 5         | 98             |
| 2               |      | Ingesting the interview dataset provided by the industry partner, working on the Architecture Description document, and delivering a first prototype that successfully passes        | 19        | 65             | 39        | 93             |
| 3               |      | Creating a basic demo to be shown to Dr. Kaiser. Starting with simple views, database and LLM communication                                                                          | 22        | 43             | 22        | 54             |
| 4               |      | Creating the first version of the frontend, finishing up the basic demo of the backend. Adding a Plan Build Process                                                                  | 10        | 33             | 13        | 32             |
| 5               |      | Closing up all features from previous sprints, finalizing the frontend so that we have a working demo                                                                                | 19        | 14             | 19        | 19             |
| 6               |      | Working on the thematic analysis logic for the mid project release                                                                                                                   | 14        | 0              | 0         | 0              |
|                 |      |                                                                                                                                                                                      |           |                |           |                |
|                 |      |                                                                                                                                                                                      |           |                |           |                |
|                 |      |                                                                                                                                                                                      |           |                |           |                |
|                 |      |                                                                                                                                                                                      |           |                |           |                |
|                 |      |                                                                                                                                                                                      |           |                |           |                |
|                 |      |                                                                                                                                                                                      |           |                |           |                |
| <b>Features</b> |      |                                                                                                                                                                                      |           |                |           |                |
| 1               |      | <b>Meeting with industry partner, designing the logo, setting up the project</b>                                                                                                     |           |                |           |                |
|                 |      | Designing Project Logo                                                                                                                                                               | 5         |                | 5         |                |
|                 |      |                                                                                                                                                                                      |           |                |           |                |
|                 |      |                                                                                                                                                                                      |           |                |           |                |
|                 |      |                                                                                                                                                                                      |           |                |           |                |
| 2               |      | <b>Ingesting the interview dataset provided by the industry partner, working on the Architecture Description document, and delivering a first prototype that successfully passes</b> |           |                |           |                |
|                 |      | Select Open source model for Generation Phase                                                                                                                                        | 3         |                | 5         |                |
|                 |      | Create PDF of software architecture description                                                                                                                                      | 8         |                | 13        |                |
|                 |      | Software Bill of Material                                                                                                                                                            | 8         |                | 13        |                |
|                 |      | Setting up the Repo with FastAPI, docker, etc                                                                                                                                        | 8         |                | 8         |                |
| 3               |      | <b>Creating a basic demo to be shown to Dr. Kaiser. Starting with simple views, database and LLM communication</b>                                                                   |           |                |           |                |
|                 |      | Simple Data persistence Layer                                                                                                                                                        | 5         |                | 5         |                |
|                 |      | View Theme Overview                                                                                                                                                                  | 8         |                | 8         |                |
|                 |      | Automated Database structure Setup                                                                                                                                                   | 1         |                | 1         |                |
|                 |      | First LLM Request Setup                                                                                                                                                              | 3         |                | 3         |                |
|                 |      | Plan Build Process                                                                                                                                                                   | 5         |                | 5         |                |

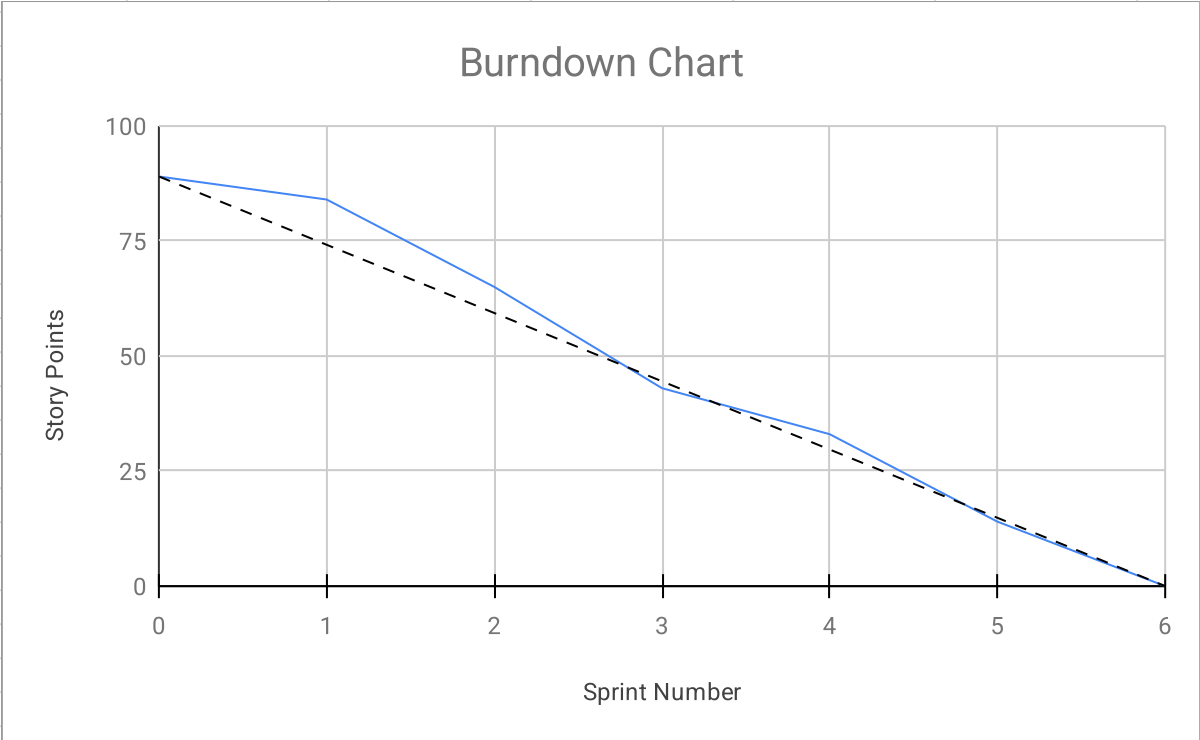
| Sprint | Goal                                                                                                                                               | Feature Name                            | Est. size | Est. remaining | Real size | Real remaining |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|-----------|----------------|-----------|----------------|
| 4      | <b>Creating the first version of the frontend, finishing up the basic demo of the backend. Adding a Readme and writting a proper Build script.</b> |                                         |           |                |           |                |
|        |                                                                                                                                                    | Upload Interview Transcript             | 3         |                | 3         |                |
|        |                                                                                                                                                    | Apply Codebook to interview             | 3         |                | 3         |                |
|        |                                                                                                                                                    | Add root <a href="#">README.md</a> file | 1         |                | 1         |                |
|        |                                                                                                                                                    | Implement Build Proccess                | 1         |                | 3         |                |
|        |                                                                                                                                                    | Setup Frontend for document Upload      | 2         |                | 3         |                |
|        |                                                                                                                                                    |                                         |           |                |           |                |
| 5      | <b>Closing up all features from previous sprints, finalizing the frontend so that we have a working demo for Industry Partner.</b>                 |                                         |           |                |           |                |
|        |                                                                                                                                                    | Setup Frontend                          | 8         |                | 8         |                |
|        |                                                                                                                                                    | Link Transcripts and Demographic Data   | 5         |                | 3         |                |
|        |                                                                                                                                                    | Upload Predefined Codebook              | 3         |                | 3         |                |
|        |                                                                                                                                                    | Upload Demographic Data                 | 3         |                | 5         |                |
|        |                                                                                                                                                    |                                         |           |                |           |                |
| 6      | <b>Working on the thematic analysis logic for the mid project release</b>                                                                          |                                         |           |                |           |                |
|        |                                                                                                                                                    | Implement Automatic Codebook Generation | 5         |                |           |                |
|        |                                                                                                                                                    | Demographic Data Upload Frontend        | 2         |                |           |                |
|        |                                                                                                                                                    | Demographic Data Viewer Page            | 2         |                |           |                |
|        |                                                                                                                                                    | Select Coding Mode Frontend             | 3         |                |           |                |
|        |                                                                                                                                                    | Export Codebook                         | 2         |                |           |                |



Scrum Master Charts







Speed

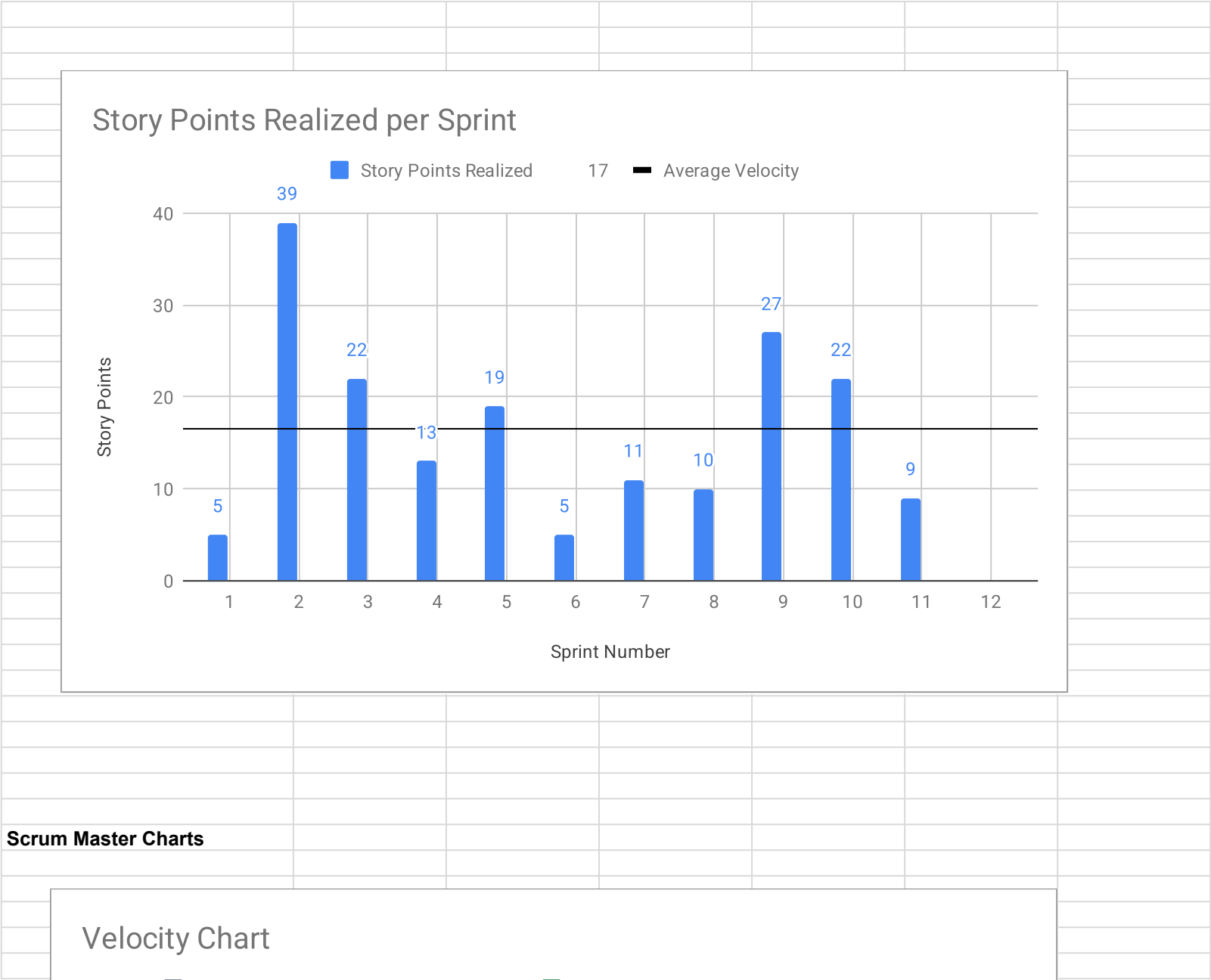
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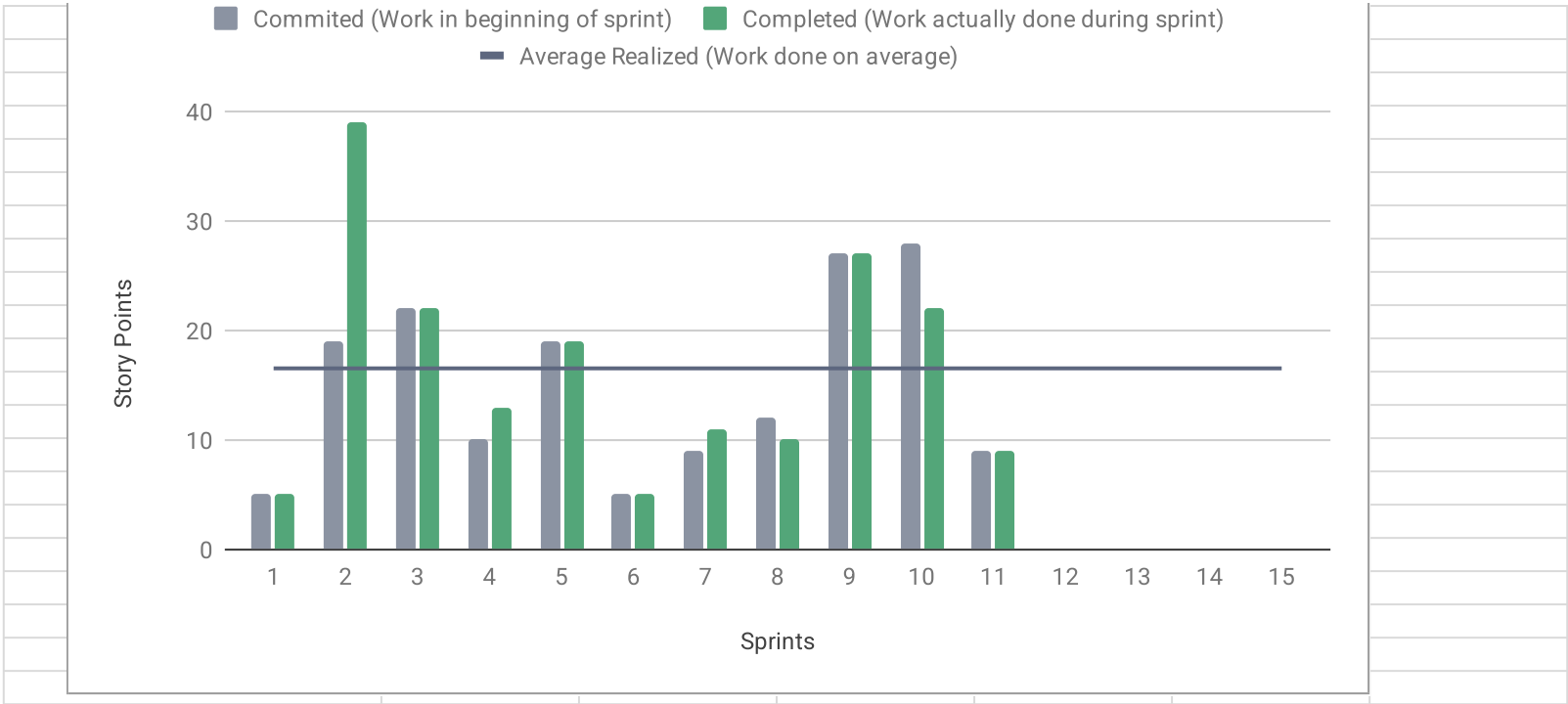
| Sprint          | Goal | Feature Name                                                                                                                                                                       | Est. size | Est. remaining | Real size | Real remaining |
|-----------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------|-----------|----------------|
| <b>Release</b>  |      |                                                                                                                                                                                    |           |                |           |                |
| <b>Total</b>    |      |                                                                                                                                                                                    | 165       |                | 182       |                |
| <b>Sprints</b>  |      |                                                                                                                                                                                    |           |                |           |                |
| 0               |      |                                                                                                                                                                                    |           |                |           |                |
| 1               |      | Meeting with industry partner, designing the logo, setting up the project                                                                                                          | 5         | 160            | 5         | 177            |
| 2               |      | Ingesting the interview dataset provided by the industry partner, working on the Architecture Description document, and delivering a first prototype that successfully pass        | 19        | 141            | 39        | 138            |
| 3               |      | Creating a basic demo to be shown to Dr. Kaiser. Starting with simple views, database and LLM communication                                                                        | 22        | 119            | 22        | 116            |
| 4               |      | Creating the first version of the frontend, finishing up the basic demo of the backend. Adding a FastAPI layer                                                                     | 10        | 109            | 13        | 103            |
| 5               |      | Closing up all features from previous sprints, finalizing the frontend so that we have a working demo                                                                              | 19        | 90             | 19        | 84             |
| 6               |      | Working on the thematic analysis logic for the mid project release                                                                                                                 | 5         | 85             | 5         | 79             |
| 7               |      | Finishing work of Sprint 6. Refactoring and Optimization                                                                                                                           | 9         | 76             | 11        | 68             |
| 8               |      | Start on Apply codebook to Corpus                                                                                                                                                  | 12        | 64             | 10        | 58             |
| 9               |      | Finish Apply Codebook to Corpus and other minor issues                                                                                                                             | 27        | 37             | 27        | 31             |
| 10              |      | Focus on displaying all of the transcripts for a particular theme and adding external API Support                                                                                  | 28        | 9              | 22        | 9              |
| 11              |      | Finish Analysis Visualization, Integrate incremental Codebook Generation and Application                                                                                           | 9         | 0              | 9         | 0              |
| <b>Features</b> |      |                                                                                                                                                                                    |           |                |           |                |
| 1               |      | <b>Meeting with industry partner, designing the logo, setting up the project</b>                                                                                                   |           |                |           |                |
|                 |      | Designing Project Logo                                                                                                                                                             | 5         |                | 5         |                |
| 2               |      | <b>Ingesting the interview dataset provided by the industry partner, working on the Architecture Description document, and delivering a first prototype that successfully pass</b> |           |                |           |                |
|                 |      | Select Open source model for Generation Phase                                                                                                                                      | 3         |                | 5         |                |
|                 |      | Create PDF of software architecture description                                                                                                                                    | 8         |                | 13        |                |
|                 |      | Software Bill of Material                                                                                                                                                          | 8         |                | 13        |                |
|                 |      | Setting up the Repo with FastAPI, docker, etc                                                                                                                                      | 8         |                | 8         |                |
| 3               |      | <b>Creating a basic demo to be shown to Dr. Kaiser. Starting with simple views, database and LLM communication</b>                                                                 |           |                |           |                |
|                 |      | Simple Data persistence Layer                                                                                                                                                      | 5         |                | 5         |                |
|                 |      | View Theme Overview                                                                                                                                                                | 8         |                | 8         |                |
|                 |      | Automated Database structure Setup                                                                                                                                                 | 1         |                | 1         |                |
|                 |      | First LLM Request Setup                                                                                                                                                            | 3         |                | 3         |                |
|                 |      | Plan Build Process                                                                                                                                                                 | 5         |                | 5         |                |

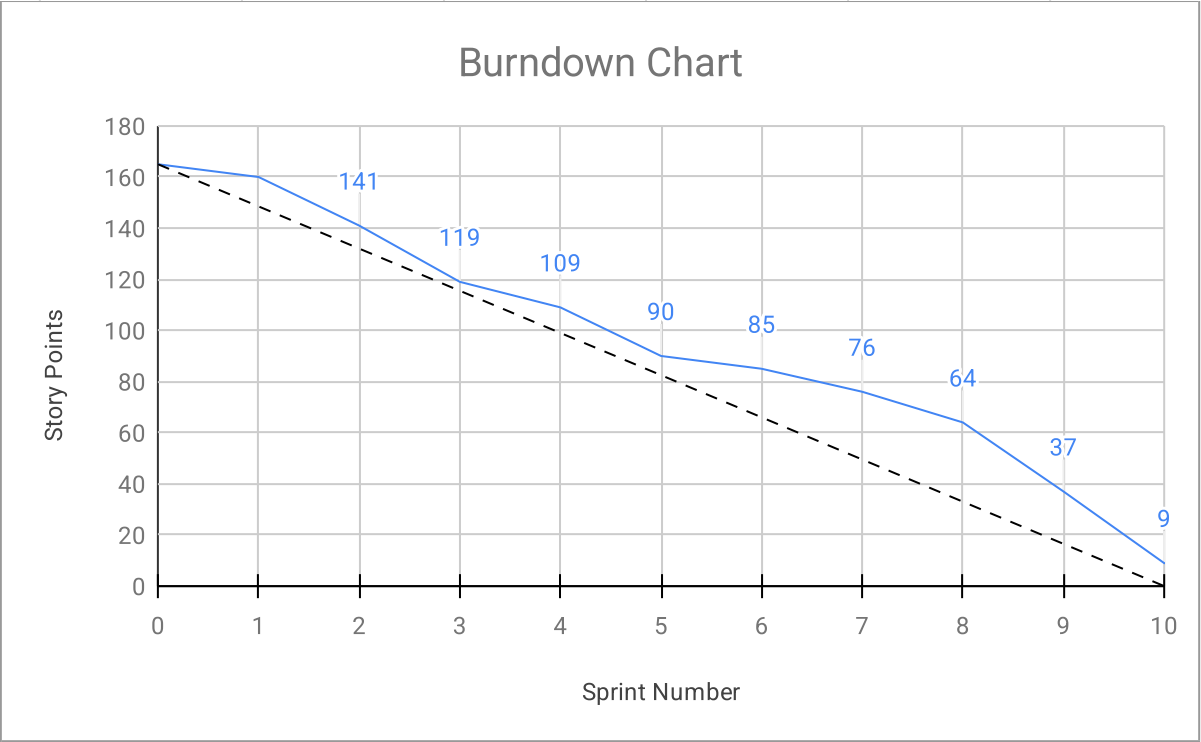
| Sprint | Goal                                                                                                                                               | Feature Name                                     | Est. size | Est. remaining | Real size | Real remaining |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|-----------|----------------|-----------|----------------|
| 4      | <b>Creating the first version of the frontend, finishing up the basic demo of the backend. Adding a Readme and writting a proper Build script.</b> |                                                  |           |                |           |                |
|        |                                                                                                                                                    | Upload Interview Transcript                      | 3         |                | 3         |                |
|        |                                                                                                                                                    | Apply Codebook to interview                      | 3         |                | 3         |                |
|        |                                                                                                                                                    | Add root <a href="#">README.md</a> file          | 1         |                | 1         |                |
|        |                                                                                                                                                    | Implement Build Process                          | 1         |                | 3         |                |
|        |                                                                                                                                                    | Setup Frontend for document Upload               | 2         |                | 3         |                |
| 5      | <b>Closing up all features from previous sprints, finalizing the frontend so that we have a working demo for Industry Partner.</b>                 |                                                  |           |                |           |                |
|        |                                                                                                                                                    | Setup Frontend                                   | 8         |                | 8         |                |
|        |                                                                                                                                                    | Link Transcripts and Demographic Data            | 5         |                | 3         |                |
|        |                                                                                                                                                    | Upload Predefined Codebook                       | 3         |                | 3         |                |
|        |                                                                                                                                                    | Upload Demographic Data                          | 3         |                | 5         |                |
| 6      | <b>Working on the thematic analysis logic for the mid project release</b>                                                                          |                                                  |           |                |           |                |
|        |                                                                                                                                                    | Implement Automatic Codebook Generation          | 5         |                | 5         |                |
| 7      | <b>Finishing work of Sprint 6. Refactoring and Optimization</b>                                                                                    |                                                  |           |                |           |                |
|        |                                                                                                                                                    | Demographic Data Upload Frontend                 | 2         |                | 2         |                |
|        |                                                                                                                                                    | Demographic Data Viewer Page                     | 2         |                | 2         |                |
|        |                                                                                                                                                    | Select Coding Mode Frontend                      | 3         |                | 3         |                |
|        |                                                                                                                                                    | Export Codebook                                  | 2         |                | 2         |                |
|        |                                                                                                                                                    | Automatic Codebook Frontend                      | 2         |                | 2         |                |
| 8      | <b>Start on Apply codebook to Corpus</b>                                                                                                           |                                                  |           |                |           |                |
|        |                                                                                                                                                    | Delete Transkript                                | 3         |                | 2         |                |
|        |                                                                                                                                                    | Delete Demographic Data                          | 3         |                | 2         |                |
|        |                                                                                                                                                    | Delete Codebook                                  | 3         |                | 3         |                |
|        |                                                                                                                                                    | Enter Natural Language Query                     | 3         |                | 3         |                |
| 9      | <b>Finish Apply Codeboook to Corpus and other minor issues</b>                                                                                     |                                                  |           |                |           |                |
|        |                                                                                                                                                    | Apply Codebook to Corpus                         | 8         |                | 8         |                |
|        |                                                                                                                                                    | Semi-Automatic Codebook Creation                 | 3         |                | 3         |                |
|        |                                                                                                                                                    | Linking Interviews and Demographic Data manually | 5         |                | 5         |                |
|        |                                                                                                                                                    | Delete Corpus                                    | 3         |                | 3         |                |
|        |                                                                                                                                                    | View Interviewee Demographics                    | 3         |                | 3         |                |
|        |                                                                                                                                                    | Read Full Transcript                             | 3         |                | 3         |                |
|        |                                                                                                                                                    | Trigger Codebook Application                     | 2         |                | 2         |                |
| 10     | <b>Focus on displaying all of the transcripts for a particular theme and adding external API Support. Minor fixes requested by Dr.Kaiser</b>       |                                                  |           |                |           |                |
|        |                                                                                                                                                    | Export Analysis Results                          | 5         |                | 5         |                |
|        |                                                                                                                                                    | LLM API Configuration UI                         | 3         |                | 1         |                |



| Sprint | Goal | Feature Name                                                                             | Est. size | Est. remaining | Real size | Real remaining |
|--------|------|------------------------------------------------------------------------------------------|-----------|----------------|-----------|----------------|
|        |      | LLM API Configuration Backend                                                            | 3         |                | 5         |                |
|        |      | Finish Codebook Application and Updating Frontend                                        | 5         |                | 3         |                |
|        |      | Add Generate Codebook Button to Upload Page                                              | 2         |                | 1         |                |
|        |      | Add Theme Description to Codebook Page                                                   | 2         |                | 2         |                |
|        |      | Drill Down into a Theme                                                                  | 8         |                | 5         |                |
|        |      |                                                                                          |           |                |           |                |
| 11     |      | Finish Analysis Visualization, Integrate incremental Codebook Generation and Application |           |                |           |                |
|        |      | Highlight Themes on Interview View                                                       | 3         |                | 3         |                |
|        |      | Choose Analysis Runs on Codebook View Page                                               | 2         |                | 2         |                |
|        |      | View Analsys Button                                                                      | 2         |                | 2         |                |
|        |      | Remove "Code" from Codebook Editing Page                                                 |           |                |           |                |
|        |      | Delete Analysis Runs                                                                     | 2         |                | 2         |                |







Speed

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| # | Feature Definition of Done                                                         | Sprint Release Definition of Done                                                           | Project Release Definition of Done                                                |
|---|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
|   | New feature code has been documented, READMEs have been updated to reflect changes | Project builds, deploys, and tests successfully                                             | Deployment process is tested and documented                                       |
|   | Unit tests for relevant logic are written and passing                              | All stories committed as “done” meet the Feature Definition of Done                         | User documentation has been updated                                               |
|   | Feature has been fully implemented                                                 | All completed features are integrated into the main branch                                  | Architecture documentation and Bill of Materials is up to date                    |
|   | Feature has been merged into the mainline                                          | Documentation and READMEs reflect changes made in the sprint                                | The released product increment solves the intended user problem at a usable level |
|   | All acceptance criteria were met                                                   | Incomplete items are clearly identified and moved/replanned                                 | Performance is acceptable for the expected usage volume                           |
|   | Product owner approved features                                                    | Retrospective completed                                                                     | The user is able to install and use the application with minimal configuration    |
|   | Developers agreed to release                                                       | All implemented features are moved to the feature archive and their issues marked as closed | No critical or high bugs remain                                                   |
|   |                                                                                    |                                                                                             |                                                                                   |
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| Type | Link / reference |
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| #  | Context                        | Name          | Version     | License                                                              | Comment                                                                                                                                               |
|----|--------------------------------|---------------|-------------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Database                       | PostgreSQL    | 18          | PostgreSQL License                                                   | Reliable open-source relational database for structured application data.                                                                             |
| 2  | Database extension             | pgvector      | 0.8         | PostgreSQL License                                                   | Adds vector storage and similarity search directly inside PostgreSQL                                                                                  |
| 3  | Programming language / runtime | Python        | 3.11        | Python Software Foundation License Version 2                         | Core runtime for backend                                                                                                                              |
| 4  | Database ORM / SQL toolkit     | SQLAlchemy    | 2           | MIT                                                                  | Provides Python ORM and SQL toolkit support for database access and queries.                                                                          |
| 5  | Backend API framework          | FastAPI       | 0.136       | MIT                                                                  | Build easy python based APIs                                                                                                                          |
| 6  | Data validation                | Pydantic      | 2.13        | MIT                                                                  | Validates, parses, and serializes Python data models safely.                                                                                          |
| 7  | LLM application framework      | LangChain     | 1.2         | MIT                                                                  | Build LLM workflows                                                                                                                                   |
| 8  | NLP library                    | spaCy         | 3.8         | MIT                                                                  | Provides production-ready NLP features such as tokenization, NER, and pipelines. Contains e.g. BERT Implementations for NLP tasks like classification |
| 9  | ASGI server                    | Uvicorn       | 0.46        | BSD-3-Clause                                                         | Runs the FastAPI backend                                                                                                                              |
| 10 | PostgreSQL async driver        | asyncpg       | 0.31        | Apache License 2.0                                                   | Provides fast asynchronous PostgreSQL connectivity for Python services.                                                                               |
| 11 | LLM observability              | Langfuse      | 4.5         | MIT Expat for core; enterprise directories under separate ee/LICENSE | Tracks LLM traces, prompts, latency, costs, and evaluation data.                                                                                      |
| 12 | HTTP client                    | HTTPX         | 0.28        | BSD-3-Clause                                                         | Provides sync and async HTTP calls to model APIs and backend services.                                                                                |
| 13 | Testing framework              | pytest        | 9           | MIT                                                                  | Automated Testing                                                                                                                                     |
| 14 | Environment configuration      | python-dotenv | 1.2         | BSD-3-Clause                                                         | Loads local environment variables from .env files during development.                                                                                 |
| 15 | Tokenization                   | tiktoken      | 0.12        | MIT                                                                  | Counts and encodes tokens. Used by LangChain for Textsplitting.                                                                                       |
| 16 | LLM frontend prototyping       | Streamlit     | 1.56        | Apache License 2.0                                                   | Quickly builds Python-based web apps, dashboards, and chat prototypes for LLM applications. Used for prototyping & early previews.                    |
| 17 | Large language model           | Gemma         | Gemma 4 31B | Apache License 2.0                                                   | Candidate self-hosted open model. More details in the model selection document. Final model decision is still outstanding.                            |

| #  | Context                            | Name          | Version         | License                                                                            | Comment                                                                                                                                                                         |
|----|------------------------------------|---------------|-----------------|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 18 | Large language model               | Qwen          | Qwen3.5-35B-A3B | Apache License 2.0                                                                 | Candidate self-hosted open model. More details in the model selection document. Final model decision is still outstanding.                                                      |
| 19 | Large language model               | Mixtral       | Mixtral 8x7B    | Apache License 2.0                                                                 | Candidate self-hosted open model. More details in the model selection document. Final model decision is still outstanding.                                                      |
| 20 | LLM gateway / provider abstraction | LiteLLM       | 1.83            | MIT for OSS package; enterprise package/features under separate commercial license | Provides a unified OpenAI-compatible interface for calling multiple LLM providers.                                                                                              |
| 21 | Containerization / deployment      | Docker Engine | 29              | Apache License 2.0                                                                 | Packages and runs the application stack consistently across environments.                                                                                                       |
| 22 | Web framework                      | Flask         | 3.1             | BSD-3-Clause                                                                       | Server-side web framework for the production frontend                                                                                                                           |
| 23 | Template engine                    | Jinja2        | 3.1             | BSD-3-Clause                                                                       | Template engine used with Flask to generate server-rendered HTML views                                                                                                          |
| 24 | Chat / users                       | OpenWebUI     | 0.9             | Open WebUI License                                                                 | chat-style interaction and user management. Current OpenWebUI versions use the Open WebUI License, which is based on BSD-3-Clause but adds a branding / attribution restriction |
|    |                                    |               |                 |                                                                                    |                                                                                                                                                                                 |
|    |                                    |               |                 |                                                                                    |                                                                                                                                                                                 |



|                                                                            | First Name | Value |  |      |                  |  |  |
|----------------------------------------------------------------------------|------------|-------|--|------|------------------|--|--|
| Kahvecioglu                                                                | Selin      | 2     |  | 2.00 | OK               |  |  |
| de Sa Filho                                                                | Fabiano    | N/A   |  |      |                  |  |  |
| Ammann                                                                     | Paul       | 2     |  |      |                  |  |  |
|                                                                            |            |       |  |      |                  |  |  |
| Wendlandt                                                                  | Julian     | 2     |  | 0    | No size          |  |  |
| Müermann                                                                   | Ole        | 2     |  | 1    | Trivial size     |  |  |
| Klarer                                                                     | Paul       | 2     |  | 2    | Small size       |  |  |
|                                                                            |            |       |  | 3    | Medium size      |  |  |
| Annus                                                                      | Mohammad   | 2     |  | 5    | Large size       |  |  |
| Rasul                                                                      | Hamid      | 2     |  | 8    | Very large size  |  |  |
|                                                                            |            |       |  | 13   | Too large (size) |  |  |
|                                                                            |            |       |  |      |                  |  |  |
| How to play planning poker                                                 |            |       |  |      |                  |  |  |
|                                                                            |            |       |  |      |                  |  |  |
| 1. Everyone type their number into their value field, don't hit return yet |            |       |  |      |                  |  |  |
| 2. Someone, perhaps a product owner, count down 3.. 2.. 1..                |            |       |  |      |                  |  |  |
| 3. Then, everyone hit return to submit their value                         |            |       |  |      |                  |  |  |
|                                                                            |            |       |  |      |                  |  |  |